Weiyang "Frank" Wang

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flasew
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Education

2022 – 2026 **Massachusetts Institute of Technology**, *Cambridge*, MA
Ph.D. Computer Science
Advisor: Prof. Manya Ghobadi
Thesis: Workload-Driven Network Optimizations for Distributed Machine Learning
Minor in Quantum Computing and Quantum Information Science

2020 – 2022 **Massachusetts Institute of Technology**, *Cambridge*, MA
M.S. Computer Science
Advisor: Prof. Manya Ghobadi
Thesis: TopoOpt: Co-optimizing Network Topology and Parallelization Strategy for Distributed Machine Learning Training Jobs

2016 – 2020 **University of California, San Diego**, *La Jolla*, CA, *magna cum laude*
B.S. Computer Science, *Honors With Highest Distinction*
Advisor: Prof. Alex C. Snoeren
Thesis: Harnessing Highly Dynamic Datacenter Fabrics with TDTCP
B.S. Physics
Minor in Mathematics

Research Interest

I design and build adaptive network systems that harness structures in machine learning workloads to enable network-application co-optimization.

Publications

SIGCOMM 2026 **MonkeyTree: Near-Minimal Congestion for Multi-tenant Training via Migration**, Anton A. Zabreyko, **Weiyang Wang**, Manya Ghobadi, *Proceedings of the ACM SIGCOMM 2026 Conference, Denver, CO, August 2026*

OSDI 2026 **PeeR: First-Class Scheduling for Latency Critical eBPF Applications**, Jeremy Carin, Ben Holmes, **Weiyang Wang**, Ankit Bhardwaj, Manya Ghobadi, *Proceedings of the 20th USENIX Symposium on Operating Systems Design and Implementation (OSDI), Seattle, WA, July 2026*

OFC 2026 **Reconfiguration-Aware Direct-Connect AI Cluster using Spatial-and-Wavelength-Selective Switching**, Brett George, **Weiyang Wang**, Zhenguo Wu, Yuyang Wang, Xiang Meng, Manya Ghobadi, Keren Bergman, *Optical Fiber Communication Conference (OFC) 2026*
Top-Scored paper

NSDI 2026 **Checkmate: Zero Performance Overhead Model Checkpointing via Network Gradient Replication**, Ankit Bhardwaj*, **Weiyang Wang***, Jeremy Carin, Adam Belay, Manya Ghobadi (*: equal contribution), *Proceedings of the 23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI), Renton, WA, May 2026*

ICLR 2026 **KramaBench: A Benchmark for AI Systems on Data-to-Insight Pipelines over Data Lakes**, Eugenie Lai*, Gerardo Vitagliano*, Ziyu Zhang*, Om Chabra, Sivaprasad Sudhir, Anna Zeng, Anton A. Zabreyko, Chenning Li, Ferdi Kossmann, Jialin Ding, Jun Chen, Markos Markakis, Matthew Russo, **Weiyang Wang**, Ziniu Wu, Michael J. Cafarella, Lei Cao, Samuel Madden, Tim Kraska (*: equal contribution), *Proceedings of the International Conference on Learning Representations (ICLR) 2026, Rio de Janeiro, Brazil, April 2026*

IEEE Micro 2025 **Spine-Free Networks for LLM Training**, **Weiyang Wang**, Manya Ghobadi, *IEEE Micro, vol. 45, no. 2, pp. 18-25, March-April 2025*

NSDI 2025 **Efficient Direct-Connect Topologies for Collective Communications**, Liangyu Zhao, Siddharth Pal, Tapan Chugh, **Weiyang Wang**, Prithwish Basu, Joud Khoury, Arvind Krishnamurthy, *Proceedings of the 22nd USENIX Symposium on Networked Systems Design and Implementation (NSDI), Philadelphia, PA, April 2025*

HotI 2024 **Rail-only: A Low-Cost High-Performance Network for Training LLMs with Trillion Parameters**, **Weiyang Wang**, Manya Ghobadi, Kayvon Shakeri, Ying Zhang, Naader Hasani, *Proceedings of IEEE Symposium on High-Performance Interconnects (HOTI), Online Conference, August 2024*
Supported by major network device vendors like Juniper and Broadcom

NSDI 2023 **TopoOpt: Co-optimizing Network Topology and Parallelization Strategy for Distributed Training Jobs**, **Weiyang Wang**, Moein Khazraee, Zhizhen Zhong, Zhihao Jia, Dheevatsa Mudigere, Ying Zhang, Anthony Kewitsch, and Manya Ghobadi, *Proceedings of the 20th USENIX Symposium on Networked Systems Design and Implementation (NSDI), Boston, MA, April 2023*

SIGCOMM 2022 **Time-division TCP for Reconfigurable Data Center Networks**, Shawn Shuoshuo Chen*, **Weiyang Wang***, Christopher Canel, Srinivasan Seshan, Alex C. Snoeren, Peter Steenkiste (*: equal contribution), *Proceedings of the ACM SIGCOMM 2022, Amsterdam, Netherlands, August 2022*

OptSys 2021 **IOI: In-network Optical Inference**, Zhizhen Zhong, **Weiyang Wang**, Manya Ghobadi, Alexander Sludds, Ryan Hamerly, Liane Bernstein, Dirk Englund, *Proceedings of the ACM SIGCOMM 2021 Workshop on Optical Systems (OptSys), Online Conference, August 2021*

NSDI 2020 **Adapting TCP for Reconfigurable Datacenter Networks**, Matthew Mukerjee, Christopher Canel, **Weiyang Wang**, Daehyeok Kim, Srinivasan Seshan, and Alex C. Snoeren, *Proceedings of the 17th USENIX Symposium on Networked Systems Design and Implementation (NSDI), Santa Clara, CA, February 2020*

Honors and Awards

Feb 2022 **Finalist**, *Meta PhD Research Fellowship Program in Computer Networks*

Sep 2020 **Awardee**, *MIT Presidential Fellowship Award*

Jun 2020 **Awards of Excellence**, *UCSD Jacobs School of Engineering, Award for Excellence in Computer Science and Engineering*

Dec 2019 **Runners-Up**, *CRA Outstanding Undergraduate Researcher Awards*

Dec 2018 **Winner**, *UCSD CSE Department, Computer Networking Espresso Prize*

Dec 2016 – Jun 2020 **Provost Honors**, *UCSD*

Press Releases

Aug 2024 **This AI Network Has No Spine – And That's A Good Thing**, *The Next Platform*

Apr 2023 **Meta, MIT, Others Test Robotic Arm in Optical AI Infrastructure**, *HPC Wire*

Apr 2023 **Telescent and MIT CSAIL Collaborate to Accelerate Machine Learning Workflows**, *Business Wire*

Posters

Apr 2025 **Efficient Direct-Connect Topologies for Collective Communications**, Liangyu Zhao, Siddharth Pal, Tapan Chugh, **Weiyang Wang**, Jason Fantl, Prithwish Basu, and Joud Khoury, Arvind Krishnamurthy, *Poster Session at the 22nd USENIX Symposium on Networked Systems Design and Implementation (NSDI), Philadelphia, PA, April 2025*

Apr 2024 **Zero Buffer Optical Packet Switching Data Center Network**, Shawn Shuoshuo Chen, **Weiyang Wang**, Manya Ghobadi, Srinivasan Seshan, Peter Steenkiste, *Poster Session at the 21st USENIX Symposium on Networked Systems Design and Implementation (NSDI), Santa Clara, CA, April 2024*

Research Experiences

Mar 2023 – Now **Student Research Scholar**, *Center for Ubiquitous Connectivity (CUbiC), SRC JUMP 2.0 Program*
Center Leader: Prof. Keren Bergman, Columbia University

Sep 2020 – Now **Graduate Research Assistant**, *MIT Computer Science and Artificial Intelligence Lab*
Advisor: Prof. Manya Ghobadi

Jan 2019 – Jun 2020 **Undergraduate Research Assistant**, *UCSD CSE Department*
Advisor: Prof. Alex C. Snoeren

Jun 2018 – Sep 2018 **Undergraduate Researcher**, *UCSD CSE Department*
Mentor: Prof. Yannis Papakonstantinou

Feb 2017 – Jun 2018 **Undergraduate Researcher**, *UCSD Center of Astrophysics and Space Science*
Mentors: Dr. Praween Siritanasak, Alex Zahn, and Prof. Brian Keating

Industry Experiences

May 2024 – Aug 2024 **Research Intern**, *Microsoft Research*, Improving Speed and Robustness of ML Training
Mentor: Dr. Srikanth Kandula

Jun 2023 – Aug 2023 **Research Intern**, *Google Cloud*, Topology Engineering and Traffic Engineering under Demand Uncertainty
Mentor: Dr. Anny Xijia Zheng

Teaching Experiences

Fall 2024 **Guest Lecturer**, *MIT EECS Department*, Computer Networks (6.5820)
Delivered a full lecture on distributed machine learning

Fall 2023 **Guest Lecturer**, *MIT EECS Department*, Computer Networks (6.5820)
Delivered part of a lecture on networks for machine learning

Fall 2022 **Teaching Assistant**, *MIT EECS Department*, Computer Networks (6.5820)

Winter 2020 **Tutor**, *UCSD CSE Department*, Introduction to Programming II (CSE8B)

Fall 2018 – Fall 2019 **Tutor**, *UCSD CSE Department*, Database Principles (CSE132A)

Mentoring Experiences

Fall 2024 – Now **Jeremy Carin**, *Ph.D. Student with Prof. Manya Ghobadi, MIT CSAIL*
With Jeremy, we explored the mismatch of increasing functionalities of today's eBPF applications and Linux's existing eBPF implementation. This research has led to submissions to HotNets 2025 and OSDI 2026.

Fall 2023 – Now **Anton Zabreyko**, *Ph.D. Student with Prof. Manya Ghobadi, MIT CSAIL*
With Anton, we are investigating network optimization that combines placement, routing, and flow scheduling in multitenant ML training clusters.

Spring 2024 – Now **Om Chabra**, *Ph.D. Student with Prof. Hari Balakrishnan, MIT CSAIL*
I provided project feedback and writing guidance to Om on his projects about training ML models in space, and distributed fallback networks.

Spring 2023 **Natalie Muradyan**, *Undergraduate Research Opportunities Program (UROP), B.S. Computer Science and Engineering, MIT*
Natalie developed a demonstration website of TopoOpt with me. She went on to pursue an M.Eng. with the Microarchitecture ATtacks and CHAllenges (MATCHA) research group at MIT with Prof. Mengjia Yan.

Invited Talks

A Tale of Two Networks: Defining Network Infrastructure for Deep Neural Network Training

Oct 2025 Computer Science Colloquia, Tufts University, *Medford, MA*
Mar 2025 ELEN E9403: Seminar in Photonics, Columbia University, *New York, NY*
Jan 2024 Networking Lecture Series, Microsoft Research, *Online*

Reconfigurable Network Architecture for DNN Training

Jun 2025 Workshop on Reconfigurable Networks, Cornell University, *Ithaca, NY*
May 2024 SRC JUMP 2.0 CUbiC Liaison Presentation, *Online*

Zero-Overhead Model Checkpointing via Network Gradient Replication

Apr 2025 Distributed Systems Laboratory (DSL) Seminar, University of Pennsylvania, *Online*

Rail-only: A Low-Cost High-Performance Network for Training LLMs with Trillion Parameters

Aug 2025 MIT / Jane Street Research Symposium, Jane Street Capital, *New York, NY*

Sep 2024 HPC Applications, SW, and HW Sync Seminar, Advanced Micro Devices, Inc. (AMD), *Online*

Aug 2024 MLSys Seminar, University of Washington, *Seattle, WA*

TopoOpt: Co-optimizing Network Topology and Parallelization Strategy for Distributed Training Jobs

Feb 2023 Network Research Group Seminar, Carnegie Mellon University, *Pittsburgh, PA*

Oct 2022 S2Infra Talks, Google Inc., *Online*

Dec 2021 MSR Cambridge, Lecture Series, Microsoft Research, *Online*

Jan 2021 Internal Presentation, Telescent Inc., *Online*

Patent

Jul 2024 **In-network Optical Inference**, Manya Ghobadi, Zhizhen Zhong, **Weiyang Wang**, Liane Bernstein, Alexander Sludds, Ryan Hamerly, Dirk Robert Englund, *US Patent Application Number 18561985*

Professional Services

2026 **Program Committee**, *IEEE Symposium on High-Performance Interconnects (HotI)*

2026 **Program Committee**, *Workshop on Hot Topics in Optical Technologies and Applications in Networking (HotOptics)*

2026 **Program Committee**, *ACM Symposium on Cloud Computing (SoCC)*

2026 **Program Committee**, *ACM Asia-Pacific Workshop on Networking (APNet)*

2025 **Reviewer**, *IEEE Network Magazine*

2025 **Reviewer**, *IEEE/ACM Transactions on Networking (ToN)*

2025 **Reviewer**, *IEEE Open Journal of the Solid-State Circuits Society (OJ-SSCS)*

2025 **Reviewer**, *Elsevier Computer Networks (COMNET)*

2025 **External Reviewer**, *IEEE International Conference on Network Protocols (ICNP)*

2025 **External Reviewer**, *ACM Asia-Pacific Workshop on Networking (APNet)*

2025 **External Reviewer**, *IEEE Global Communications Conference (GLOBECOM)*

Other Services

Sep 2022 – Jan 2026 **Member**, *MIT EECS REFS*
Served as a peer mediator to support the graduate community and serve as a first point of contact in dealing with stress

Sep 2022 – May 2026 **Web Chair**, *Sidney-Pacific Student Government, MIT*
Served as Web Chair to maintain and develop the administration system for MIT's Sidney-Pacific graduate housing

Sep 2020 – May 2026 **System Administrator**, *Network and Mobile Systems Research Group, MIT*
Constructed and managed a 24-node GPU cluster, with electrical and optical networking, from scratch